

SVKM'S NMIMS

MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT & ENGINEERING/
SCHOOL OF TECHNOLOGY MANAGEMENT

B.Tech/MBA Tech/B.T. Academic Year: 2024-2025

Programs: ~~B.Tech~~ Stream: All applicable streams / Mecha / Com

Year: III/V Semester: V/IX

Subject: Elements of Project Management

Marks: 100

No. of Pages: 09

Synoptic

Final Examination (2024-25)

Q1	Answer briefly:	[20]
a.	Importance of Project management in industries - Construction: - Controls costs and timelines, avoiding delays and overruns. - Manages risks (weather, accidents) and ensures quality compliance. - Optimizes resource allocation (labor, materials, equipment). - Manufacturing: - Ensures smooth production flow and reduces downtime. - Manages supply chains and schedules to avoid bottlenecks. - Improves product quality through continuous monitoring. - IT: - Keeps projects on time and within scope (e.g., software development). - Manages changing requirements and risks effectively. - Enhances team collaboration through agile methods and task tracking. In all industries, project management ensures efficiency, risk control, and successful project delivery.	[5]
b.	The project manager is the centerpiece of project management because they coordinate every aspect of the project, ensuring alignment among tasks, timelines, resources, and stakeholders. They: - Plan and Organize – - Lead and Motivate Teams – - Monitor Progress – - Ensure Delivery – - Communicate with Stakeholders –. Without a skilled project manager, even well-designed projects can face delays, scope creep, or failure due to poor coordination and decision-making. <i>Explain Any relevant five points</i>	[5]

	c.	Benefits of Responsibility Assignment Matrix: 1. Clear Role Definition: 2. Improved Communication: 3. Accountability: 4. Efficient Resource Allocation: 5. Risk Reduction: 6. Enhanced Decision-Making: The RAM ensures smoother project execution by aligning tasks with the right people, reducing miscommunication, and increasing accountability.	<i>List any five relevant benefits</i>	[5]
	d.	Importance of information monitoring and control in project risk management: 1. Early Risk Detection: 2. Informed Decision-Making: 3. Prevents Scope Creep: 4. Ensures Accountability: 5. Improves Stakeholder Communication: 6. Enhances Adaptability: Effective information monitoring and control ensures risks are managed efficiently, minimizing disruptions and increasing the likelihood of project success.	<i>any five.</i>	[5]
Q2				[20]
	a.	1. Scope: <i>(Any five points)</i> a. Defines the boundaries, objectives, deliverables, and tasks of a project. b. Prevents scope creep by setting clear expectations about what is included and excluded. c. Ensures alignment among stakeholders and provides a roadmap for execution. d. Helps determine the manpower, equipment, and materials required to complete the project. e. Provides a baseline for tracking progress and measuring project success against predefined objectives. 2. Feasibility: <i>(Any five points)</i> a. Assesses whether the project is realistically achievable within time, budget, and resource constraints. b. Evaluates technical, financial, and operational viability to avoid project failures. c. Ensures risks are identified early and helps determine whether to proceed with the project.		[10]

		d. Evaluates potential obstacles, ensuring risks are managed proactively. e. Ensures the project aligns with business goals and stakeholder expectations, avoiding conflicts later.	
	b.	Role of organisation structures (5M) 1. Defines Authority and Responsibility: Clarifies roles, decision-making authority, and reporting lines. 2. Facilitates Communication: Establishes channels for information flow between teams, departments, and stakeholders. 3. Resource Allocation: Determines how resources (personnel, budget, tools) are shared across projects and departments. 4. Influences Project Execution: Affects how quickly decisions are made and how flexible the project team can be (e.g., matrix vs. functional structure). 5. Affects Accountability: Determines who is responsible for project outcomes—project manager, functional managers, or both. organisation structures (5M)	[10]
Q.3			[20]
	a.	Steps for construction of WBS: 1. Define Project Objectives: 2. Identify Major Deliverables: 3. Break Down Deliverables into Subtasks: 4. Organize the WBS Hierarchically: 5. Assign Responsibilities: 6. Review and Validate the WBS: 7. Maintain the WBS: Example: (5M) 1. Residential Building Project 1.1 Site Preparation 1.1.1 Land Clearing 1.1.2 Excavation 1.1.3 Grading 1.2 Foundation 1.2.1 Footing 1.2.2 Slab 1.2.3 Waterproofing 1.3 Structure 1.4 Roofing 1.5 Electrical	[10]

		1.6 Plumbing	
	b.	• Steps Involved in the Cost Estimating Process for a Project: 1. Define Project Scope: 2. Identify Cost Components: 3. Gather Data: 4. Choose Estimation Techniques: 5. Develop the Cost Estimate: 6. Review and Validate Estimates: 7. Document Assumptions and Risks: 8. Create a Cost Baseline: • Uses of Cost Estimates in Project Management: Budgeting: • Provides a financial framework to allocate resources effectively and ensure project feasibility. • Enables tracking of actual costs against estimates, helping to identify variances and implement corrective actions. Any four (8 M) (2 M)	
Q4			[20]
	a.	Risk analysis is the process of identifying, assessing, and prioritizing risks associated with a project to minimize their impact on objectives. It helps project managers understand the potential challenges and develop strategies to mitigate them. Methods of Risk Analysis: (2 M) 1. Qualitative Risk Analysis (4 M) • Description: Involves assessing risks based on their probability of occurrence and potential impact on project objectives. • Techniques: ○ Risk Matrix: ○ Expert Judgment: ○ SWOT Analysis: 2. Quantitative Risk Analysis (4 M) • Description: Involves numerically estimating the effects of identified risks on project objectives, often through statistical methods. • Techniques: ○ Monte Carlo Simulation: ○ Decision Tree Analysis: ○ Sensitivity Analysis:	[10]

	b. Key Deliverables in Project Management (6M) (3M) ○ Final Project Report: A comprehensive document summarizing the entire project, including objectives, scope, methodologies, outcomes, and lessons learned. Provides stakeholders with an overview of project performance, successes, challenges, and recommendations for future projects. (3M) ○ Post-Implementation Review (PIR): An evaluation conducted after project completion to assess whether the project met its objectives and to identify areas for improvement. Gathers feedback from stakeholders, reviews project effectiveness, and informs future projects by analyzing what worked well and what didn't. Challenges During Project Handover: (4M) ○ Knowledge Transfer: ○ Stakeholder Alignment: ○ Resource Availability: ○ Change Management: ○ Documentation Completeness: ○ Integration with Existing Systems: Any four.	[10]
Q5		
	a. Role of Gantt Charts in Planning and Scheduling: (4M) • Visual Representation: • Task Tracking: • Dependency Management: <i>Explain Any two</i> Advantages: 1. Clarity and Simplicity: 2. Improved Planning: 3. Enhanced Coordination: 4. Progress Monitoring: <i>Any three</i> (3M) Disadvantages: 1. Limited Detail: 2. Static Nature: 3. Dependency Complexity: 4. Time-Consuming: <i>Any three</i> (3M)	[10]

	b.	Challenges Encountered in Managing the Budget: (5M) • Unforeseen Expenses: • Scope Changes (Scope Creep): • Inaccurate Estimations: • Resource Allocation Issues: • Poor Communication: Techniques to Ensure the Project Stayed Within Budget: (5M) • Detailed Budget Planning: • Regular Monitoring and Reporting: • Change Control Procedures: • Contingency Reserves: • Use of Project Management Software:	[10]
Q6			
	a.	Project Management is often defined by the triple constraints: scope, time, and cost. These constraints are interrelated, meaning that a change in one will impact the others. (4M) • Scope: • Time: • Cost: Challenges in Balancing the Triple Constraints: (3M) • Scope Creep: • Time Constraints: • Budget Overruns: Strategies to Address Challenges: (3M) • Change Control Process: • Regular Monitoring and Adjustments: • Stakeholder Communication:	[10]

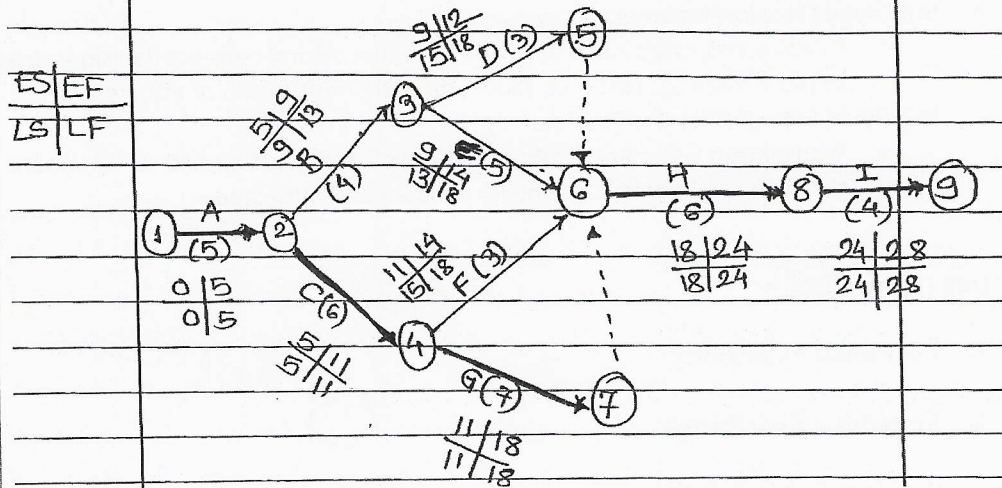
	b. Process of Monitoring and Control in Project Risk Management: (6M) 1. Risk Identification: 2. Risk Analysis: 3. Risk Response Planning: 4. Monitoring Risks: 5. Reviewing Risk Responses: 6. Reporting: Impact on Project Performance and Cost Control (4M) • Proactive Management: ○ By monitoring and controlling risks, project managers can anticipate and address issues before they escalate, minimizing disruptions to project timelines and quality. • Cost Control: ○ Effective risk management helps prevent cost overruns by identifying potential financial impacts early, allowing for budget adjustments and contingency planning. • Improved Decision-Making: ○ Ongoing risk assessments provide data that informs project decisions, leading to more strategic resource allocation and prioritization of efforts. • Enhanced Stakeholder Confidence: ○ Transparent risk management processes foster trust among stakeholders, as they can see how risks are being handled and mitigated.	[10]
Q7		
	a. Types of Feasibility (4M) 1. Technical Feasibility: 2. Economic Feasibility: 3. Operational Feasibility: 4. Legal Feasibility: 5. Schedule Feasibility: Challenges Faced During Feasibility Studies: (4M) 1. Incomplete Data: 2. Changing Requirements: 3. Stakeholder Disagreement: 4. Resource Limitations: 5. Risk Underestimation:	[4] [8]

b

$$\frac{T_o + 4T_m + T_p}{6} = \frac{3 + 4(5) + 7}{6} = 5$$

$$\left(\frac{T_p - T_o}{6} \right)^2 = \left(\frac{7 - 3}{6} \right)^2 = 0.4444$$

A	P	T _o	T _m	T _p	T _E	σ^2
A	—	3	5	7	5	0.4444
B	A	2	4	6	4	0.4444
C	A	4	6	8	6	0.4444
D	B	1	3	5	3	0.4444
E	B	3	5	7	5	0.4444
F	C	2	3	4	3	0.1111
G	C	5	7	9	7	0.4444
H	D, E, F, G	4	6	8	6	0.4444
I	H	2	4	6	4	0.4444



Activity	Predecessors	TE	Variance
A	None	5.0	0.444444
B	A	4.0	0.444444
C	A	6.0	0.444444
D	B	3.0	0.444444
E	B	5.0	0.444444
F	C	3.0	0.111111
G	C	7.0	0.444444
H	D, E, F, G	6.0	0.444444
I	H	4.0	0.444444

Critical Path: A → C → G → H → I

Total Duration: 28 days

Activities on the critical path:

- C: Variance = 0.444444
- G: Variance = 0.444444
- H: Variance = 0.444444
- I: Variance = 0.444444

$$\text{Total Variance} = 0.444444 + 0.444444 + 0.444444 + 0.444444 = 1.777776$$

$$\text{Standard Deviation}(\sigma) = \sqrt{1.777776} \approx 1.33 \quad (2M)$$

Next, we can use the Z-score formula to find the probability:

$$Z = \frac{X - \mu}{\sigma} = \frac{25 - 28}{1.33} \approx -2.26$$

Using standard normal distribution tables or calculators, a Z-score of -2.26 corresponds to a probability of approximately 0.0116 or 1.16%. (2M)

